Chapter 11 - Inheritance & More in OOPs

Inheritance is a way of creating a new class from an existing class.

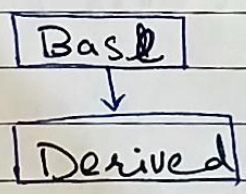
Syntax :

```
Class Employee :        # Base class
    # code
class Programmer(Employee) :   # Derived or
    #code.                      child class
```

- We can use the method and attributes of 'Employee' in 'Programmer' object.
- Also, we can overwrite or add new methods in 'Programmer' class.

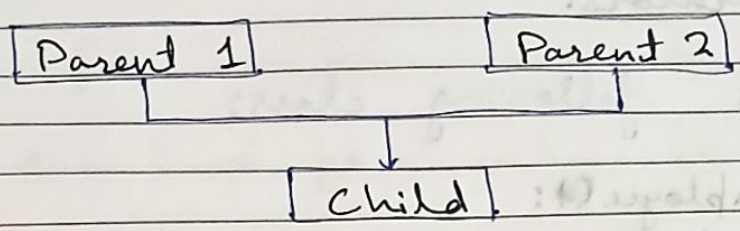
Types of Inheritance :
- Single inheritance
- Multiple inheritance
- Multilevel inheritance.

Single Inheritance :
Single inheritance occurs when child class inherits only a single parent class.



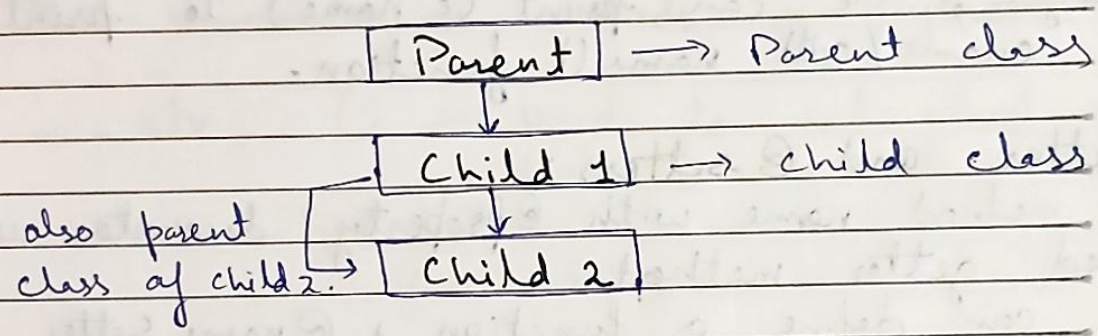
Multiple Inheritance :

Multiple inheritance occurs when the child class inherits from more than one parent classes.



Multilevel Inheritance :

When a child class becomes a parent of another child class.



Super() Method :

Super method is used to access the methods of a super class in the derived class.

```
Super().__init__()
    # __init__() calls constructor of the base class.
```

Class Method :

A class method is a method which is bound to the class and not the object of the class.

@classmethod decorator is used to create class method.

Syntax :

```
@Classmethod
    def (cls ,p1 ,p2) :
```

Property decorators:

Consider the following class:

```
Class Employee:
    @property
    def name(self):
        return self.ename.
```

if e = Employee() is an object of class employee, we can print (e.name) to print the ename / call name() function.

@.getter and @.setters :

The method name with @property decorator is called getter method.
We can define a function + @name.setter decorator like below:

```
@name.setter
def name(self ,value):
    self.ename = value.
```

Operator overloading in Python.

Operators in python can be overloaded using dunder methods.
These methods are called when a given operator is used on the objects.

Operators in python can be overloaded using the following methods:

p1 + p2 → p1 __add__(p2)

p1 - p2 → p1 __sub__(p2)

p1 * p2 → p1 __mul__(p2)

p1 / p2 → p1 __truediv__(p2)

p1 // p2 → p1 __floordiv__(p2)

Other dunder/magic methods in python.

__str__() → used to set what gets displayed upon calling str(obj)

__len__() → used to set what gets displayed upon calling __len__() or len(obj)